

HI-ER Readme

This code is used for the <https://er.pumps.org/> website.

Getting the Application

The application consists of many folders and files. In order to allow multiple people to work on the HI-ER content, and for continued application development, the HI-ER is managed through a [git](#) repository. [git](#) is a platform that allows for very powerful version control - it tracks changes and allows you to "commit" them and "push" your changes to a central repository so others can always see the latest work.

Git needs to be installed on your machine. You can download it [here](#)

We will not be using many features of [git](#), and since the number of people editing the HI-ER content will be somewhat limited, it will be rare for you to need any advanced knowledge of how git works.

The initial step is to create a Github account - this website is the central repository for the HI-ER, and it is a private repository - you will need to be granted access before moving forward. Create your account [here](#), and contact me (sfrees@intelliquip.com) for access.

Once you have access, open the Command Prompt and use the `cd` command to navigate to the directory you want to put the HI-ER application in. For example, create a directory called `C:\projects\`, and navigate the command prompt there by entering `cd C:\projects` at the command prompt.

Next, clone the repository with the following command:

```
git clone https://github.com/edlpumps/hi-er.git
```

This will create a directory called `hi-er` under your `C:\projects` directory.

Later, as you begin to make changes to the HI-ER, you will need to use [git](#) to commit your changes. The details of this is covered below in the "Version control with [git](#)" section.

Text Editor - Visual Studio Code

While any text editor is suitable for creating the HI-ER content, you should use something geared towards programming, to avoid character encoding problems. You may download and install this [here](#).

Once installed, you can open the `hi-er` directory.

Visual Studio Code allows you to open a folder - `hi-er` - which is the most efficient way of working. This will give you a side panel on the left side of the screen that you can use to navigate and open any file in the directory structure.

Running code Locally

Initialize Node JS and NPM in your Development environment

Follow instructions at: <https://code.visualstudio.com/docs/nodejs/nodejs-tutorial>

Additional information can be found at: <https://developer.ibm.com/tutorials/learn-nodejs-installing-node-nvm-and-vscode/>

Confirm npm is installed

```
$> npm --help
```

Confirm node.js is installed

```
$> node --version
```

Just one time, install the modules for your development environment:

In your VSC app and in a **powershell** terminal window:

```
$> npm init  
$> npm install  
$> npm audit fix <== if needed
```

Local Database

You must be connected to a local Mongoose database install MongoDB install MongoDB Compass create .env with:

- MONGO_CONNECTION_DATA = mongodb://127.0.0.1:27017/er
- ADMIN_PASSWORD_OVERRIDE=test

You will need to have your user information in the database.

If MongoDB Compass is running but it won't connect to the local database:

- Open Task Manager
- Click on the Services tab
- Make sure MongoDB Server is running

Local Preview - Debugging

The easiest way to do your development is to execute the following command from the root project directory in a Terminal window:

```
$> node index.js
```

OR to set breakpoints and debug:

```
Open index.js file in VSC
Click on the Debug icon on the left
Select Launch Program
Step through code
Use the debug toolbar to step into/over code
Use the debug toolbar to restart the code and stop execution
```

To monitor for code changes while running index.js, install **supervisor**:

```
$>npm init
$>npm install
$>npm -g install supervisor
```

Then run the code using **supervisor**:

```
$>supervisor index.js
```

Connect to the localhost on port:

```
http://127.0.0.1:3003
```

Deploying a GitHub branch to Heroku

First, push the branch to GitHub and create a Pull Request.

Login to Heroku using the **higladetech@gmail.com** account on a browser:

```
https://heroku.com
Select the "intelliquip-hi/hi-er-beta" project (make sure that it is NOT in
Maintenance mode in the Settings)
Select Deploy
Select Deploy a GitHub branch and enter the branch name
```

Deploying GitHub Master branch to Heroku

Make sure all branches are merged to **master** on GitHub.

Follow the steps above to Deploy a GitHub branch to Heroku, but instead, select the **intelliquip-hi/hi-er** project and deploy the **master** branch.

ER LABELS

- There are **hard coded** values for Pump Annual Run Hours (4000), Circulator Annual Run Hours (2500) and the Cost/KwH (\$0.15). These are used in the Energy & Cost Savings calculations.
([routes/common.js](#))
- The year of the DOE Circulator Efficiency Regulation is a variable that is passed into the label pub
([utils/label_builder.js](#), [views/svg/circulator-label.pug](#))
- These values can easily become Environment Variables.
- The *Meets the 2028 DOE circulator efficiency regulation* line on the large *Circulator Pump Label* should only display if the CEI value that is displayed in the black bar on the label is less than or equal to 1.00.
([utils/label_builder.js](#))

PNG to URI

- If you add an image to the Energy Ratings label (such as a logo or icon), you need to convert the image to URI content and save that as a file.
 - Use an online coverter like <https://site24x7.com/tools/image-to-datauri.html>
 - Upload the image (PNG, JPEG, whatever) to the tool which generates the URI code.
 - Copy the content and paste into a file (no extension) in the views/svg folder
 - Pass that link into the SVG generator code for generating the label.
- The URI that is generated for the HI logos on the large label are generated from:
 - English ([views/svg/hi-title](#)): hydraulic-institute-logo.png
 - French ([views/svg/hi-title-fr](#)): HI-Energy-Rating-ID_French_03_Transparent.png
- The URI that is generated for the HI logos on the small label are generated from:
 - English ([views/svg/hi-title-small](#)): hydraulic-institute-logo-sm.png
 - French ([views/svg/hi-title-small-fr](#)):
HI-Energy-Rating-ID_French_03_small_Transparent.png
- The URI that is generated for the Circulator Labels with the approval check on the bottom of the large label are generated from:
 - All languages ([views/svg/hi-approval-check](#)): HI-Approval-Check-s.png

TRANSLATIONS TO OTHER LANGUAGES

- Pump LABELS can be translated using the English/French toggle at the top right of all Pump Detail pages.
- The pump details themselves are only translated for the PUBLIC pages (not the Admin or Participant pages per design).
- The i18next (<https://i18next.com>) library is used.
- Since the requirement was to provide translations for particular pages (Public Pump details) and for labels, there are 2 "types" of language settings.
 - page_lang
 - label_lang
- These are handled independently in the code.

GOOGLE ANALYTICS

- The Google Analytics ID is G-1RZZNP84J1. Code is in global_includes.jade

TESTING SUBSCRIBER EMAIL ATTACHMENTS (FULL & QPL LISTINGS)

- In your development environment, to test the generation of the subscriber Full and QPL spreadsheets, run `export-file.js`:

```
$> node export-file.js
```

Generate README.PDF

- Make sure the `Markdown PDF` extension is installed on Visual Studio Code (Author yzane).
- Open the README.md file
- Press the 'F1' key
- Select the `Markdown PDF` extension (or search for it). The README.pdf file will automatically be generated.

Copying MongoDB Collections between databases

- Use `mongodump` and `mongorestore` commands from a terminal window.
- Example: Copy all collections from production to local into the 'er' database:

```
$> mongodump --uri="mongodb://hi-db-admin:<password>@hi-er-prod-shard-00-00.kq34k.mongodb.net:27017,hi-er-prod-shard-00-01.kq34k.mongodb.net:27017,hi-er-prod-shard-00-02.kq34k.mongodb.net:27017/heroku_b5n55m48?authSource=admin&replicaSet=atlas-obx42v-shard-0&w=majority&readPreference=primary&retryWrites=true&ssl=true" --out ~/dump
```

```
$> mongorestore --host localhost -db er ~/dump/heroku_b5n55m48
```